ANALYYSI I: LIMITS

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1. Course information

The official course information is available on the University of Helsinki course page.

2. Lecture notes

2.1. **01.09.2026.**

2.1.1. *The absolute value function.*

Definition 2.1. For any real number $x \in \mathbb{R}$, the *absolute value* (or *modulus*) of x, denoted by $|x|$, is defined by

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0. \end{cases}$$

Geometrically, $|x - y|$ is the distance between the points x and y on the real line.

Example 2.2. Fix $y \in \mathbb{R}$ and let $h > 0$. The inequality $|x - y| < h$ can be written in several equivalent ways:

$$\begin{aligned} |x - y| < h &\iff -h < x - y < h \\ &\iff y - h < x < y + h \\ &\iff x \in (y - h, y + h). \end{aligned}$$

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Thus, $|x-y|<h$ means that x lies in the open interval of radius h centred at y.

Example 2.3. As a concrete instance of the preceding equivalences,

$$\begin{aligned} |2x-1|<3 &\iff -3<2x-1<3 \\ &\iff -2<2x<4 \\ &\iff -1<x<2 \\ &\iff x\in(-1,2). \end{aligned}$$

Theorem 2.4 (Triangle and reverse triangle inequalities)**.** *For all $x,y\in\mathbb{R}$,*

$$\begin{aligned} |x+y|&\le|x|+|y|, \\ ||x|-|y||&\le|x-y|. \end{aligned}$$

Proof. By definition 2.1,

$$\begin{aligned} -|x|\le x\le|x|, \\ -|y|\le y\le|y|. \end{aligned}$$

Adding the inequalities gives

$$-\big(|x|+|y|\big)\le x+y\le|x|+|y|,$$

which is equivalent to $|x+y|\le|x|+|y|$.

Applying this triangle inequality to $x=(x-y)+y$ gives

$$|x|-|y|\le|x-y|.$$

Interchanging x and y similarly gives $|y|-|x|\le|x-y|$. Therefore

$$-|x-y|\le|x|-|y|\le|x-y|,$$

which is equivalent to the reverse triangle inequality. □

2.2. **03.09.2026.**

2.2.1. *More on absolute values.*

Theorem 2.5. *We have*

$$|xy|=|x||y|$$

for all $x,y\in\mathbb{R}$.

Proof. We prove by splitting into four cases. Firstly, if $x,y\ge0$, we have $xy\ge0$, so $|xy|=xy$. But in this case also both $|x|=x$ and $|y|=y$, so $|xy|=xy=|x||y|$. Secondly, if $x,y<0$, we have $xy>0$, $|xy|=xy=(-x)(-y)=|x||y|$. Now suppose $x<0$ and $y\ge0$. Then $xy\le0$, and hence $|xy|=-xy=(-x)y=|x||y|$. The case $x\ge0$, $y<0$ follows from commutativity of multiplication of real numbers. □

Example 2.6. Let $x\in(1,3)$ and $y\in(4,8)$. Find an upper bound for

$$|x+y-8|.$$

Kolmioepäyhtälön nojalla saadaan

$$\begin{aligned} |x+y-8| &= |(x-2)+(y-6)| \\ &\le |x-2|+|y-6| \\ &< 1+2 \\ &= 3. \end{aligned}$$

Thus we have found that 3 is an upper bound for $|x+y-8|$.

Example 2.7. Let $x \in (2,6)$ and $y \in (2,8)$. Find an upper boud for

$$|xy - 20|.$$

We have $|x-4| < 2$ and $|y-5| < 3$. We have

$$\begin{aligned} |xy-20| &= |xy - 4y + 4y - 20| \\ &\leq |xy - 4y| + |4y - 20| \\ &= |y||x-4| + 4|y-5| \\ &\leq 8 \cdot 2 + 4 \cdot 3 \\ &= 28. \end{aligned}$$

It follows that 28 is an upper bound for $|xy - 20|$.

Example 2.8. Find a positive real number m such that

$$|x^2 - 16| \leq m|x-4|,$$

where $x \in (0,10)$.

When $x = 4$, we get $0 \leq 0$, which holds with every $m \in \mathbb{R}$. Now suppose $x \neq 4$. We simplify the inequality and obtain

$$|x+4| \leq m.$$

Since $x \in (0,10)$, $|x+4| \in (4,14)$. We have thus obtained

$$|x+4| \leq 14,$$

so we can choose $m = 14$.

2.2.2. *Field and ordering properties of* $\mathbb{R}$.

Definition 2.9. $(F, +, \cdot)$ is a field if and only if

(1) The algebraic structure $(F, +)$ is an *abelian group*.
(2) The algebraic structure $(F^*, \cdot)$ is an *abelian group*, where $F^* = F \setminus \{0_F\}$.
(3) The operation $\cdot$ distributes over $+$.

Theorem 2.10. $\mathbb{R}$ *is a field.*

Proof. Consider the usual addition and multiplication on $\mathbb{R}$.

First, $(\mathbb{R}, +)$ is an abelian group. Addition is closed, associative, and commutative; 0 is an additive identity; and every $x \in \mathbb{R}$ has the additive inverse $-x$.

Next, $(\mathbb{R} \setminus \{0\}, \cdot)$ is an abelian group. Multiplication is associative and commutative, and 1 is its identity element. If $x \neq 0$, then $x^{-1} = 1/x$ is a nonzero real number satisfying

$$x \cdot x^{-1} = 1.$$

The product of two nonzero real numbers is nonzero, so multiplication is closed on $\mathbb{R} \setminus \{0\}$.

Finally, multiplication distributes over addition:

$$x(y+z) = xy + xz$$

for all $x, y, z \in \mathbb{R}$. Therefore $(\mathbb{R}, +, \cdot)$ is a field. □

Definition 2.11 (Ordered field)**.** An *ordered field* is a field $(F, +, \cdot)$ equipped with a relation $<$ such that, for all $x, y, z \in F$,

(1) exactly one of $x < y$, $x = y$, or $x > y$ holds;
(2) if $x < y$ and $y < z$, then $x < z$;
(3) if $x < y$, then $x + z < y + z$; and
(4) if $0_F < x$ and $0_F < y$, then $0_F < xy$.

Here $x > y$ means $y < x$. The first two conditions state that $<$ is a strict total order, while the last two state that the order is compatible with the field operations.

Theorem 2.12. *The field $(\mathbb{R}, +, \cdot)$, equipped with the usual order $<$, is an ordered field.*

Proof. We have already shown that $(\mathbb{R}, +, \cdot)$ is a field. The usual strict order on $\mathbb{R}$ satisfies trichotomy and transitivity, so it satisfies the first two ordering axioms.

Suppose that $x < y$. Then $0 < y - x$, and for every $z \in \mathbb{R}$,

$$(y + z) - (x + z) = y - x > 0.$$

Hence $x + z < y + z$. Furthermore, if $0 < x$ and $0 < y$, then the usual sign rule for multiplication gives $0 < xy$. Thus the order is compatible with addition and multiplication, and $\mathbb{R}$ is an ordered field. □

Theorem 2.13 (Uniqueness of additive identity)**.** *The identity element 0_F of the abelian group $(F, +)$ is unique.*

Proof. Suppose that 0_F and $0'_F$ are both additive identity elements of $(F, +)$. Since 0_F is an additive identity,

$$0_F + 0'_F = 0'_F.$$

Since $0'_F$ is also an additive identity,

$$0_F + 0'_F = 0_F.$$

Therefore $0_F = 0'_F$. □

Theorem 2.14 (Cancellation law)**.** *Let $x, y, z \in \mathbb{R}$. If $x + y = x + z$, then $y = z$.*

Proof. Using the additive identity, additive inverse, associativity, and the assumption $x + y = x + z$, we obtain

$$\begin{aligned} y &= 0 + y \\ &= ((-x) + x) + y \\ &= (-x) + (x + y) \\ &= (-x) + (x + z) \\ &= ((-x) + x) + z \\ &= 0 + z \\ &= z. \end{aligned}$$

□

2.3. **08.09.2026.**

2.3.1. *Sequences.*

Definition 2.15. Let X and Y be sets. A function $f\colon X \to Y$ is a relation $f \subseteq X \times Y$ such that, for every $x \in X$, there exists exactly one $y \in Y$ with $(x, y) \in f$. This unique y is denoted by $f(x)$.

The set X is the *domain* of f, and Y is its *codomain*. The set

$$f(X) = \{f(x) : x \in X\}$$

is its *range* or *image*.

Definition 2.16. A sequence of real numbers is a function $x\colon \mathbb{N} \to \mathbb{R}$. For $n \in \mathbb{N}$, the value $x(n)$, usually denoted by x_n, is the n-th term of the sequence. We denote the sequence by $(x_n)_{n\in\mathbb{N}}$, or simply by (x_n).

2.3.2. *Limit of a sequence.*

Definition 2.17. Let (x_n) be a sequence of real numbers. We say that (x_n) *converges* to $a \in \mathbb{R}$ if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that, for every $n \in \mathbb{N}$ with $n > N$,

$$|x_n - a| < \varepsilon.$$

In this case, we write $x_n \to a$ as $n \to \infty$, or $\lim_{n\to\infty} x_n = a$.

Let us look at some examples.

Example 2.18. Let (x_n) be defined by

$$x_n = \frac{2n}{n+1}.$$

We will show that

$$\lim_{n\to\infty} x_n = 2.$$

Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > \frac{2}{\varepsilon}$. Now for every $n > N$, we have

$$\begin{aligned} |x_n - 2| &= \left|\frac{2n}{n+1} - 2\right| \\ &= \frac{2}{n+1} < \frac{2}{n} < \frac{2}{\frac{2}{\varepsilon}} = \varepsilon. \end{aligned}$$

Thus $x_n \to 2$ when $n \to \infty$.

Example 2.19. Let (x_n) be defined by

$$\begin{aligned} x_0 = \cdots = x_7 &= 42, \\ x_n &= 7 + \frac{1}{n}, \quad n \geq 8. \end{aligned}$$

We will show that

$$\lim_{n\to\infty} x_n = 7.$$

Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > \max\left\{8, \frac{1}{\varepsilon}\right\}$. Now for every $n > N$, we have

$$|x_n - 7| = \frac{1}{n} < \varepsilon.$$

Thus $x_n \to 7$ as $n \to \infty$.

Let us now tackle a simple divergence proof.

Example 2.20. Let us show that the sequence x_n defined by $x_n = n$ diverges. Assume that the sequence converged to some $a \in \mathbb{R}$. Taking $\varepsilon = 1$, there exists $N \in \mathbb{N}$ such that, for every $n > N$,

$$|x_n - a| < \varepsilon = 1.$$

Choose $n \in \mathbb{N}$ such that

$$n > \max\{N, |a| + 1\}.$$

By the triangle inequality,

$$n = |n| = |(n - a) + a| \leq |n - a| + |a|.$$

Subtracting $|a|$ from both sides and using $n > |a| + 1$, we obtain

$$|x_n - a| = |n - a| \geq n - |a| > 1,$$

contradicting the assumption $|x_n - a| < 1$. Therefore (x_n) diverges.

Example 2.21. Let (x_n) be defined by

$$x_0 = \cdots = x_{42} = 1,$$
$$x_n = \frac{n}{n-42}, \quad n \geq 43.$$

We will show that

$$\lim_{n\to\infty} x_n = 1.$$

Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that

$$N > \max\left\{84, \frac{84}{\varepsilon}\right\}.$$

Now, for every $n > N$, we have $n > 84$, and hence $\frac{n}{2} - 42 > 0$. Therefore

$$\begin{aligned} |x_n - 1| &= \left|\frac{n}{n-42} - 1\right| \\ &= \frac{42}{n-42} \\ &= \frac{42}{\frac{n}{2} + \frac{n}{2} - 42} \\ &< \frac{84}{n} < \varepsilon. \end{aligned}$$

Thus $x_n \to 1$ as $n \to \infty$.

2.4. **10.09.2026.**

2.4.1. *Uniqueness of limits.*

Theorem 2.22 (Uniqueness of limits)**.** *A convergent sequence of real numbers has a unique limit. That is, if $x_n \to a$ and $x_n \to b$ as $n \to \infty$, then $a = b$.*

Proof. Suppose for contradiction that $a \neq b$, and set

$$\varepsilon = \frac{|a-b|}{2} > 0.$$

Since $x_n \to a$ and $x_n \to b$, there exist $N_a, N_b \in \mathbb{N}$ such that

$$n > N_a \implies |x_n - a| < \varepsilon$$

and

$$n > N_b \implies |x_n - b| < \varepsilon.$$

Choose $n > \max\{N_a, N_b\}$. By the triangle inequality,

$$|a-b| \leq |a - x_n| + |x_n - b| < 2\varepsilon = |a-b|,$$

which is impossible. Therefore $a = b$. □

Theorem 2.23 (Convergent sequences are bounded)**.** *Every convergent sequence of real numbers is bounded.*

Proof. Suppose that $x_n \to a$. Taking $\varepsilon = 1$, there exists $N \in \mathbb{N}$ such that

$$n > N \implies |x_n - a| < 1.$$

By the triangle inequality, for every $n > N$,

$$|x_n| \leq |x_n - a| + |a| < 1 + |a|.$$

Let

$$M = \max\{|x_0|, |x_1|, \ldots, |x_N|, 1 + |a|\}.$$

Then $|x_n| \leq M$ for every $n \in \mathbb{N}$, so (x_n) is bounded. □

2.4.2. *Algebra of limits.*

Theorem 2.24 (Algebraic limit laws)**.** *Suppose that $x_n \to a$ and $y_n \to b$, and let $c \in \mathbb{R}$. Then*

(1) $x_n \pm y_n \to a \pm b$.
(2) $cx_n \to ca$.
(3) $x_n y_n \to ab$.
(4) If $b \neq 0$, then $y_n \neq 0$ for all sufficiently large n, and

$$\frac{x_n}{y_n} \to \frac{a}{b}.$$

Proof. (1) Let $\varepsilon > 0$. Since $x_n \to a$ and $y_n \to b$, there exist $N_x, N_y \in \mathbb{N}$ such that

$$n > N_x \implies |x_n - a| < \frac{\varepsilon}{2}$$

and

$$n > N_y \implies |y_n - b| < \frac{\varepsilon}{2}.$$

Thus, for $n > \max\{N_x, N_y\}$,

$$|(x_n + y_n) - (a + b)| \leq |x_n - a| + |y_n - b| < \varepsilon.$$

Hence $x_n + y_n \to a + b$. Replacing y_n and b by $-y_n$ and $-b$ gives $x_n - y_n \to a - b$.

(2) If $c = 0$, then $cx_n = ca = 0$ for every n. If $c \neq 0$, choose $N \in \mathbb{N}$ such that

$$n > N \implies |x_n - a| < \frac{\varepsilon}{|c|}.$$

Then, for every $n > N$,

$$|cx_n - ca| = |c||x_n - a| < \varepsilon.$$

Therefore $cx_n \to ca$.

(3) Since (y_n) converges, it is bounded. Choose $M > 0$ such that $|y_n| \leq M$ for every n. There exist $N_1, N_2 \in \mathbb{N}$ such that

$$n > N_1 \implies |x_n - a| < \frac{\varepsilon}{2M}$$

and

$$n > N_2 \implies |y_n - b| < \frac{\varepsilon}{2(|a| + 1)}.$$

For $n > \max\{N_1, N_2\}$,

$$\begin{aligned} |x_n y_n - ab| &\leq |x_n y_n - a y_n| + |a y_n - ab| \\ &= |y_n||x_n - a| + |a||y_n - b| \\ &< M\frac{\varepsilon}{2M} + |a|\frac{\varepsilon}{2(|a| + 1)} \\ &< \varepsilon. \end{aligned}$$

Therefore $x_n y_n \to ab$.

(4) Suppose that $b \neq 0$. Since $y_n \to b$, there exists $N_0 \in \mathbb{N}$ such that

$$n > N_0 \implies |y_n - b| < \frac{|b|}{2}.$$

The reverse triangle inequality then gives

$$|y_n| \geq |b| - |y_n - b| > \frac{|b|}{2} > 0$$

for every $n > N_0$. In particular, $y_n \neq 0$ eventually. Choose $N_1 \in \mathbb{N}$ such that

$$n > N_1 \implies |y_n - b| < \frac{\varepsilon |b|^2}{2}.$$

For $n > \max\{N_0, N_1\}$,

$$\left|\frac{1}{y_n} - \frac{1}{b}\right| = \frac{|y_n - b|}{|y_n||b|} < \frac{2}{|b|^2}|y_n - b| < \varepsilon.$$

Hence $1/y_n \to 1/b$; the finitely many terms with $n \leq N_0$ may be defined arbitrarily because they do not affect the limit. Applying the product law to (x_n) and $(1/y_n)$ yields

$$\frac{x_n}{y_n} \to \frac{a}{b}.$$

□

2.4.3. *Limits and order.*

Theorem 2.25. *Suppose that $x_n \to x$. Then*

(1) if $x_n \leq M$ for every $n \in \mathbb{N}$, then $x \leq M$;
(2) if $x_n \geq m$ for every $n \in \mathbb{N}$, then $x \geq m$.

Proof. (1) Suppose for contradiction that $x > M$, and set

$$\varepsilon = \frac{x - M}{2} > 0.$$

Since $x_n \to x$, there exists $N \in \mathbb{N}$ such that, for every $n > N$,

$$|x_n - x| < \varepsilon.$$

It follows that

$$x_n > x - \varepsilon = x - \frac{x - M}{2} = \frac{x + M}{2} > M,$$

contradicting $x_n \leq M$. Therefore $x \leq M$.

(2) Suppose for contradiction that $x < m$, and set

$$\varepsilon = \frac{m - x}{2} > 0.$$

Since $x_n \to x$, there exists $N \in \mathbb{N}$ such that, for every $n > N$,

$$|x_n - x| < \varepsilon.$$

It follows that

$$x_n < x + \varepsilon = x + \frac{m - x}{2} = \frac{x + m}{2} < m,$$

contradicting $x_n \geq m$. Therefore $x \geq m$.

□

Theorem 2.26 (Squeeze theorem)**.** *Suppose that*

$$x_n \leq y_n \leq z_n$$

for every $n \in \mathbb{N}$. If $x_n \to a$ and $z_n \to a$, then $y_n \to a$.

Proof. Let $\varepsilon > 0$. Since $x_n \to a$, there exists $N_x \in \mathbb{N}$ such that

$$n > N_x \implies a - \varepsilon < x_n < a + \varepsilon.$$

Similarly, since $z_n \to a$, there exists $N_z \in \mathbb{N}$ such that

$$n > N_z \implies a - \varepsilon < z_n < a + \varepsilon.$$

Therefore, for every $n > \max\{N_x, N_z\}$,

$$a - \varepsilon < x_n \le y_n \le z_n < a + \varepsilon.$$

Hence $|y_n - a| < \varepsilon$, so $y_n \to a$. □

The inequalities in the squeeze theorem need only hold for all sufficiently large n, since changing finitely many terms does not affect convergence.

Example 2.27. Consider the oscillating sequence

$$x_n = \frac{(-1)^n}{n}, \qquad n \ge 1.$$

Its sign alternates, but

$$-\frac{1}{n} \le \frac{(-1)^n}{n} \le \frac{1}{n}.$$

Both bounding sequences converge to 0. Indeed, given $\varepsilon > 0$, choosing $N > 1/\varepsilon$ gives $1/n < \varepsilon$ whenever $n > N$. The squeeze theorem therefore gives

$$\lim_{n\to\infty} \frac{(-1)^n}{n} = 0.$$

Notice that the squeeze theorem lets us prove convergence without trying to handle even and odd indices separately.

Example 2.28. Consider

$$x_n = \sqrt{n^2 + n} - n, \qquad n \ge 1.$$

Rationalising the expression gives

$$x_n = \frac{n}{\sqrt{n^2 + n} + n}.$$

Since

$$n < \sqrt{n^2 + n} < n + 1,$$

we obtain

$$\frac{n}{2n + 1} < x_n < \frac{1}{2}.$$

By the algebraic limit laws,

$$\lim_{n\to\infty} \frac{n}{2n + 1} = \lim_{n\to\infty} \frac{1}{2 + 1/n} = \frac{1}{2}.$$

The upper bound is the constant sequence $1/2$, so both bounds converge to $1/2$. Hence the squeeze theorem yields

$$\lim_{n\to\infty} \left(\sqrt{n^2 + n} - n\right) = \frac{1}{2}.$$

2.4.4. *Subsequences.*

Definition 2.29. Let (x_n) be a sequence, and let (n_k) be a strictly increasing sequence of natural numbers:

$$n_0 < n_1 < n_2 < \cdots .$$

The sequence (x_{n_k}), obtained by selecting the terms of (x_n) whose indices are $n_0, n_1, n_2, \ldots$, is called a *subsequence* of (x_n).

Theorem 2.30. *If $x_n \to x$, then every subsequence of (x_n) converges to x.*

Proof. Let (x_{n_k}) be a subsequence of (x_n), and let $\varepsilon > 0$. Since $x_n \to x$, there exists $N \in \mathbb{N}$ such that

$$n > N \implies |x_n - x| < \varepsilon.$$

Because (n_k) is strictly increasing, $n_k \geq k$ for every $k \in \mathbb{N}$. Thus, whenever $k > N$, we also have $n_k > N$, and hence

$$|x_{n_k} - x| < \varepsilon.$$

Therefore $x_{n_k} \to x$. Since the subsequence was arbitrary, every subsequence of (x_n) converges to the same limit x. □

2.5. **15.09.2026.**

2.5.1. *Upper and lower bounds.*

Definition 2.31. Let $\varnothing \neq E \subseteq \mathbb{R}$. A number $s \in \mathbb{R}$ is an *upper bound* of E if

$$x \leq s \qquad \text{for every } x \in E.$$

If such a number exists, E is *bounded above*.

Definition 2.32. Let $\varnothing \neq E \subseteq \mathbb{R}$. A number $p \in \mathbb{R}$ is a *lower bound* of E if

$$p \leq x \qquad \text{for every } x \in E.$$

If such a number exists, E is *bounded below*.

Definition 2.33. Let $\varnothing \neq E \subseteq \mathbb{R}$ be bounded above. An upper bound s of E is the *least upper bound*, or *supremum*, of E if $s \leq u$ for every upper bound u of E. We write

$$s = \sup E.$$

Definition 2.34. Let $\varnothing \neq E \subseteq \mathbb{R}$ be bounded below. A lower bound p of E is the *greatest lower bound*, or *infimum*, of E if $p \geq \ell$ for every lower bound ℓ of E. We write

$$p = \inf E.$$

If E is not bounded above, we write $\sup E = +\infty$. Similarly, if E is not bounded below, we write $\inf E = -\infty$.

2.5.2. *The completeness axiom.*

Completeness axiom. Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

Corollary 2.35. *Every nonempty subset $E \subseteq \mathbb{R}$ that is bounded below has an infimum in $\mathbb{R}$.*

Proof. Let c be a lower bound of E, and define

$$F = \{-x : x \in E\}.$$

Since $c \leq x$ for every $x \in E$, we have $-x \leq -c$. Thus F is bounded above. It is also nonempty, so the completeness axiom gives $\sup F \in \mathbb{R}$. For every $x \in E$,

$$-x \leq \sup F \leq -c, \qquad \text{hence} \qquad x \geq -\sup F \geq c.$$

Therefore $-\sup F$ is a lower bound of E. Since the argument applies to every lower bound c of E, it is the greatest lower bound. Consequently,

$$\inf E = -\sup F \in \mathbb{R}.$$

□

2.5.3. *Maximum and properties of the supremum.*

Definition 2.36. A number s is the *greatest element*, or *maximum*, of a set E if $s \in E$ and $x \leq s$ for every $x \in E$. We write $s = \max E$.

Theorem 2.37. *Let $\varnothing \neq E \subseteq \mathbb{R}$ be bounded above. If $s = \max E$, then $s = \sup E$.*

Proof. Since $x \leq s$ for every $x \in E$, the number s is an upper bound of E. Hence $\sup E \leq s$. On the other hand, $s \in E$, so $s \leq \sup E$. Thus $s = \sup E$. □

Theorem 2.38. *The supremum of a nonempty subset of $\mathbb{R}$ that is bounded above is unique.*

Proof. Suppose that s and s' are both least upper bounds of E. Since s' is an upper bound and s is a least upper bound, $s \leq s'$. Similarly, since s is an upper bound and s' is a least upper bound, $s' \leq s$. Therefore $s = s'$. □

Theorem 2.39. *Let $\varnothing \neq E \subseteq \mathbb{R}$ be bounded above. For every $\varepsilon > 0$, there exists $x \in E$ such that*

$$\sup E - \varepsilon < x \leq \sup E.$$

Proof. Since $\sup E$ is an upper bound, $x \leq \sup E$ for every $x \in E$. Suppose, for a contradiction, that for some $\varepsilon > 0$,

$$x \leq \sup E - \varepsilon \qquad \text{for every } x \in E.$$

Then $\sup E - \varepsilon$ is an upper bound of E, but $\sup E - \varepsilon < \sup E$. This contradicts the fact that $\sup E$ is the least upper bound. Hence the claimed element x exists. □

2.5.4. *Monotone sequences.*

Definition 2.40. A real sequence (x_n) is

- *increasing* (nondecreasing) if $x_n \leq x_{n+1}$ for every $n \in \mathbb{N}$;
- *decreasing* (nonincreasing) if $x_{n+1} \leq x_n$ for every $n \in \mathbb{N}$;
- *strictly increasing* if $x_n < x_{n+1}$ for every $n \in \mathbb{N}$;
- *strictly decreasing* if $x_{n+1} < x_n$ for every $n \in \mathbb{N}$.

A sequence is *monotone* if it is increasing or decreasing, and *strictly monotone* if it is strictly increasing or strictly decreasing.

2.6. **17.09.2026.**

2.6.1. *Monotone convergence.*

Theorem 2.41 (Monotone convergence theorem)**.** *Let (x_n) be an increasing sequence of real numbers. If there is $M \in \mathbb{R}$ such that $x_n \leq M$ for every $n \in \mathbb{N}$, then (x_n) converges, and*

$$\lim_{n\to\infty} x_n = \sup\{x_n : n \in \mathbb{N}\} \leq M.$$

Proof. The set $E = \{x_n : n \in \mathbb{N}\}$ is nonempty and bounded above by M. By completeness, $s = \sup E$ exists in $\mathbb{R}$, and $s \leq M$. Let $\varepsilon > 0$. By the approximation property of the supremum, there is an index $N \in \mathbb{N}$ such that

$$s - \varepsilon < x_N \leq s.$$

Since (x_n) is increasing, for every $n > N$,

$$s - \varepsilon < x_N \leq x_n \leq s.$$

Hence $|x_n - s| = s - x_n < \varepsilon$, proving $x_n \to s$. □

Corollary 2.42 (Decreasing case)**.** *Let (x_n) be a decreasing sequence of real numbers. If there is $m \in \mathbb{R}$ such that $m \leq x_n$ for every $n \in \mathbb{N}$, then (x_n) converges, and*

$$\lim_{n\to\infty} x_n = \inf\{x_n : n \in \mathbb{N}\} \geq m.$$

Consequently, every bounded monotone real sequence converges.

Proof. The sequence $(-x_n)$ is increasing and bounded above by $-m$. By theorem 2.41,

$$-x_n \longrightarrow \sup\{-x_n : n \in \mathbb{N}\}.$$

Multiplying by -1 and using the relation between infima and suprema gives

$$x_n \longrightarrow -\sup\{-x_n : n \in \mathbb{N}\} = \inf\{x_n : n \in \mathbb{N}\} \geq m.$$

A bounded monotone sequence satisfies the hypotheses of either the increasing or the decreasing case, so it converges. □

2.6.2. *Examples of decreasing sequences.* In the following two examples the indices start at $n = 1$.

Example 2.43. Let

$$x_1 = 3, \qquad x_n = \frac{n}{n-1} \quad (n \geq 2).$$

Show that (x_n) is strictly decreasing and find its limit.

Solution. First, $x_2 = 2 < 3 = x_1$. For $n \geq 2$, all terms are positive and

$$\frac{x_{n+1}}{x_n} = \frac{(n+1)(n-1)}{n^2} = 1 - \frac{1}{n^2} < 1.$$

Thus $x_{n+1} < x_n$ for every $n \geq 1$. Also $x_1 > 1$, and

$$x_n = 1 + \frac{1}{n-1} > 1 \qquad (n \geq 2).$$

Hence the sequence is bounded below and converges by corollary 2.42. We identify its limit in two ways.
Method 1: the squeeze theorem. For $n \geq 2$, we have $n - 1 \geq n/2$, so

$$1 < x_n = 1 + \frac{1}{n-1} \leq 1 + \frac{2}{n}.$$

Both bounding sequences tend to 1, so $x_n \to 1$.
Method 2: the infimum. The number 1 is a lower bound of $E = \{x_n : n \geq 1\}$. If $a > 1$, choose an integer $n > 1 + 1/(a-1)$. Then

$$x_n < a \iff \frac{1}{n-1} < a - 1 \iff n > 1 + \frac{1}{a-1}.$$

Thus no $a > 1$ is a lower bound of E. Therefore $\inf E = 1$, and the decreasing case of the monotone convergence theorem yields $\lim_{n\to\infty} x_n = 1$. □

Example 2.44. Let

$$x_n = \frac{(n+1)^2}{n^2+1}, \qquad n \geq 1.$$

Show that (x_n) is decreasing and determine its limit.

Solution. Since every term is positive, we can compare successive terms using their ratio:

$$\begin{aligned}\frac{x_{n+1}}{x_n} &= \frac{(n+2)^2(n^2+1)}{((n+1)^2+1)(n+1)^2} \\ &= \frac{n^4+4n^3+5n^2+4n+4}{n^4+4n^3+7n^2+6n+2}.\end{aligned}$$

The denominator exceeds the numerator by

$$2n^2 + 2n - 2 = 2(n^2 + n - 1) > 0 \qquad (n \geq 1).$$

Therefore $x_{n+1}/x_n < 1$, so the sequence is strictly decreasing. Moreover,

$$x_n = 1 + \frac{2n}{n^2 + 1} > 1,$$

so it is bounded below and converges to $\inf\{x_n : n \geq 1\}$.

We now show that this infimum is 1. It is already a lower bound. Given $a > 1$, choose an integer $n > \max\{1, 2/(a-1)\}$. Then

$$x_n = 1 + \frac{2n}{n^2 + 1} < 1 + \frac{2}{n} < a.$$

Hence a is not a lower bound, and consequently

$$\lim_{n\to\infty} x_n = \inf\{x_n : n \geq 1\} = 1.$$

Alternatively, $1 < x_n < 1 + 2/n$ gives the same limit directly by the squeeze theorem. □

Remark 2.45. The restriction $n \geq 1$ matters for monotonicity in the second example. If we also define x_0 by the same formula, then $x_0 = 1 < x_1 = 2$, so the full sequence is not decreasing. Its tail starting at $n = 1$ is strictly decreasing, and its limit is still 1.

2.6.3. *The Bolzano–Weierstrass theorem.*

Theorem 2.46 (Bolzano–Weierstrass)**.** *Every bounded sequence of real numbers has a convergent subsequence.*

Proof. Let (x_n) be bounded. Choose $m, M \in \mathbb{R}$ such that $m \leq x_n \leq M$ for every $n \in \mathbb{N}$, and define the *tail suprema*

$$a_k = \sup\{x_n : n \geq k\}, \qquad k \in \mathbb{N}.$$

Each tail is nonempty and bounded, so completeness ensures that a_k exists and $m \leq a_k \leq M$. Since

$$\{x_n : n \geq k+1\} \subseteq \{x_n : n \geq k\},$$

the number a_k is an upper bound of the smaller tail as well. Therefore $a_{k+1} \leq a_k$. The sequence (a_k) is decreasing and bounded below, so corollary 2.42 gives a limit

$$a = \lim_{k\to\infty} a_k \in \mathbb{R}.$$

We construct a subsequence $(x_{n_j})_{j\geq 1}$ recursively. By the approximation property of the supremum, choose $n_1 \geq 1$ such that

$$a_1 - 1 < x_{n_1} \leq a_1.$$

Once n_j has been chosen, apply the same property to the tail starting at $n_j + 1$. There is an index $n_{j+1} \geq n_j + 1$ such that

$$a_{n_j+1} - \frac{1}{j+1} < x_{n_{j+1}} \leq a_{n_j+1}.$$

Thus the indices are strictly increasing, as required for a subsequence.

To verify convergence, observe that $a_{n_{j+1}} \leq a_{n_j+1}$ and $x_{n_{j+1}} \leq a_{n_{j+1}}$. Consequently, for every $j \geq 1$,

$$a_{n_j} - \frac{1}{j} < x_{n_j} \leq a_{n_j}.$$

For $j = 1$, the lower inequality follows from $a_{n_1} \leq a_1$; for later indices it follows from the recursive choice above. Since (a_{n_j}) is a subsequence of the convergent sequence (a_k), it tends to a. Also $1/j \to 0$, so both bounding sequences tend to a. The squeeze theorem now yields $x_{n_j} \to a$, proving the claim. □

2.7. **22.09.2026.**

2.7.1. *Bernoulli's inequality.*

Theorem 2.47 (Bernoulli's inequality)**.** *For every $x \in \mathbb{R}$ with $x > -1$ and every $n \in \mathbb{N}$,*

$$(1+x)^n \geq 1 + nx.$$

Proof. Fix $x > -1$. The cases $n = 0$ and $n = 1$ are equalities. Suppose that $(1+x)^n \geq 1+nx$ for some $n \geq 1$. Since $1 + x > 0$, multiplication preserves the inequality, and

$$\begin{aligned}(1+x)^{n+1} &= (1+x)(1+x)^n \\ &\geq (1+x)(1+nx) \\ &= 1 + (n+1)x + nx^2 \\ &\geq 1 + (n+1)x.\end{aligned}$$

The result follows by induction. If $x \neq 0$, then $nx^2 > 0$ in the induction step, so the inequality is strict for $n \geq 2$. □

2.7.2. *A sequence defining the number e.* Consider the sequence

$$x_n = \left(1 + \frac{1}{n}\right)^n, \qquad n \geq 1.$$

Lemma 2.48. *The sequence (x_n) is strictly increasing.*

Proof. All terms are positive. For $n \geq 1$,

$$\begin{aligned}\frac{x_{n+1}}{x_n} &= \frac{\left(1 + \frac{1}{n+1}\right)^{n+1}}{\left(1 + \frac{1}{n}\right)^n} \\ &= \left(\frac{1 + \frac{1}{n+1}}{1 + \frac{1}{n}}\right)^{n+1} \left(1 + \frac{1}{n}\right) \\ &= \left(\frac{n(n+2)}{(n+1)^2}\right)^{n+1} \frac{n+1}{n} \\ &= \left(1 - \frac{1}{(n+1)^2}\right)^{n+1} \frac{n+1}{n} \\ &> \left(1 - \frac{n+1}{(n+1)^2}\right) \frac{n+1}{n} = 1.\end{aligned}$$

The strict inequality follows from theorem 2.47, since $-1 < -1/(n+1)^2 < 0$ and $n+1 \geq 2$. Hence $x_{n+1} > x_n$. □

Lemma 2.49. *The sequence (x_n) is bounded above.*

Proof. We use the auxiliary sequence

$$y_n = \left(1 - \frac{1}{n}\right)^n, \qquad n \geq 2.$$

This sequence is positive and strictly increasing. Indeed, by theorem 2.47, for $n \geq 2$,

$$\begin{aligned}\frac{y_{n+1}}{y_n} &= \left(\frac{n^2}{n^2-1}\right)^n \frac{n}{n+1} \\ &= \left(1+\frac{1}{n^2-1}\right)^n \frac{n}{n+1} \\ &\geq \left(1+\frac{n}{n^2-1}\right)\frac{n}{n+1} \\ &= 1+\frac{1}{(n+1)(n^2-1)} > 1.\end{aligned}$$

Consequently, $(1/y_n)$ is decreasing. Also, for $n \geq 2$,

$$\left(1+\frac{1}{n}\right)\left(1-\frac{1}{n}\right) = 1-\frac{1}{n^2} < 1.$$

Dividing by $1-1/n > 0$ and raising to the power n gives

$$x_n < \left(1-\frac{1}{n}\right)^{-n} = \frac{1}{y_n} \leq \frac{1}{y_2} = \left(1-\frac{1}{2}\right)^{-2} = 4.$$

Finally, $x_1 = 2 < 4$, so the bound holds for every $n \geq 1$. □

By theorem 2.41, the increasing, bounded sequence (x_n) converges to a real number.

Definition 2.50. The limit

$$e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n$$

is called *Euler's number* (*Neperin luku* in Finnish).

2.8. **24.09.2026.**

2.8.1. *Sequences tending to infinity.*

Definition 2.51. A real sequence (x_n) *tends to positive infinity* if, for every $M \in \mathbb{R}$, there exists $N \in \mathbb{N}$ such that

$$x_n > M \qquad \text{whenever } n > N.$$

We write $x_n \to +\infty$, or $\lim_{n\to\infty} x_n = +\infty$.

Definition 2.52. A real sequence (x_n) *tends to negative infinity* if, for every $M \in \mathbb{R}$, there exists $N \in \mathbb{N}$ such that

$$x_n < M \qquad \text{whenever } n > N.$$

We write $x_n \to -\infty$, or $\lim_{n\to\infty} x_n = -\infty$.

Remark 2.53. Neither definition requires the sequence to be monotone. The symbols $+\infty$ and $-\infty$ are not real numbers, so these statements do not mean convergence to a real limit.

Theorem 2.54. *If $x_n \to +\infty$ or $x_n \to -\infty$, then (x_n) diverges as a sequence of real numbers.*

Proof. Suppose first that $x_n \to +\infty$, and assume for a contradiction that $x_n \to x \in \mathbb{R}$. Taking $M = x+1$ in the definition of an infinite limit gives $N_1 \in \mathbb{N}$ such that $x_n > x+1$ whenever $n > N_1$. Convergence to x, with $\varepsilon = 1$, gives $N_2 \in \mathbb{N}$ such that $|x_n - x| < 1$ whenever $n > N_2$. For $n > \max\{N_1, N_2\}$, we would have both $|x_n - x| > 1$ and $|x_n - x| < 1$, a contradiction.

If $x_n \to -\infty$, the same argument uses $M = x-1$. Eventually $x_n < x-1$, hence $|x_n - x| > 1$, again contradicting convergence to x. □

2.8.2. *A necessary condition for convergence.*

Theorem 2.55. *Let (x_n) be a convergent real sequence. For every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that*

$$|x_n - x_{n+k}| < \varepsilon \qquad \text{for all } n > N \text{ and } k \in \mathbb{N}.$$

Proof. Write $x = \lim_{n\to\infty} x_n$, and let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $|x_m - x| < \varepsilon/2$ for every $m > N$. If $n > N$ and $k \in \mathbb{N}$, then $n + k > N$, so the triangle inequality gives

$$|x_n - x_{n+k}| \leq |x_n - x| + |x_{n+k} - x| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

□

Definition 2.56. A real sequence satisfying the condition in theorem 2.55 is called a *Cauchy sequence*. Equivalently, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|x_p - x_q| < \varepsilon \qquad \text{whenever } p, q > N.$$

Indeed, take $n = \min\{p, q\}$ and $k = |p - q|$ to obtain this form from the preceding one. Conversely, set $p = n$ and $q = n + k$.

2.8.3. *Nested closed intervals.*

Lemma 2.57 (Nested interval lemma)**.** *Let $[a_n, b_n]$, $n \geq 1$, be nonempty closed intervals such that*

$$[a_{n+1}, b_{n+1}] \subseteq [a_n, b_n] \qquad (n \geq 1), \qquad \lim_{n\to\infty}(b_n - a_n) = 0.$$

Then $\bigcap_{n=1}^{\infty}[a_n, b_n]$ contains exactly one real number.

Proof. The nesting gives

$$a_n \leq a_{n+1} \leq b_{n+1} \leq b_n.$$

Thus (a_n) is increasing and (b_n) is decreasing. Both sequences lie in $[a_1, b_1]$, so the monotone convergence theorem gives real limits

$$a = \lim_{n\to\infty} a_n, \qquad b = \lim_{n\to\infty} b_n.$$

Since $a_n \leq b_n$, preservation of inequalities under limits yields $a \leq b$. Monotonicity also gives $a_n \leq a \leq b \leq b_n$ for every n, hence

$$0 \leq b - a \leq b_n - a_n.$$

The right-hand side tends to zero, so $a = b$. In particular, $a \in [a_n, b_n]$ for every n, proving existence of a point in the intersection.

For uniqueness, let $y \neq a$. Since $|y-a| > 0$ and $b_n - a_n \to 0$, choose n with $b_n - a_n < |y-a|$. If y also belonged to $[a_n, b_n]$, then $a, y \in [a_n, b_n]$ would imply $|y-a| \leq b_n - a_n$, a contradiction. Therefore y does not belong to the intersection, and

$$\bigcap_{n=1}^{\infty}[a_n, b_n] = \{a\}.$$

□

2.8.4. *Cauchy's convergence criterion.*

Theorem 2.58 (Cauchy's convergence criterion)**.** *A real sequence (x_n) converges if and only if it is a Cauchy sequence: for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that*

$$|x_n - x_{n+k}| < \varepsilon \qquad \text{for all } n > N \text{ and } k \in \mathbb{N}.$$

Proof. Necessity was proved in theorem 2.55. Conversely, suppose that (x_n) satisfies the Cauchy condition.

Step 1: boundedness. Apply the condition with $\varepsilon = 1$, and choose a corresponding $N_0 \in \mathbb{N}$. For every $m \geq N_0 + 1$, take $n = N_0 + 1$ and $k = m - (N_0 + 1)$. Then

$$|x_m - x_{N_0+1}| < 1, \qquad |x_m| < |x_{N_0+1}| + 1.$$

The finitely many earlier terms are bounded as well. In particular,

$$C = \max\{|x_0|, \ldots, |x_{N_0}|, |x_{N_0+1}| + 1\}$$

satisfies $|x_m| \leq C$ for every $m \in \mathbb{N}$.
Step 2: nested intervals from the tails. Define, for $n \geq 1$,

$$p_n = \inf\{x_m : m \geq n\}, \qquad s_n = \sup\{x_m : m \geq n\}.$$

Each tail is nonempty and bounded, so its infimum and supremum exist in $\mathbb{R}$. As the tails shrink, their infima increase and their suprema decrease:

$$p_n \leq p_{n+1} \leq s_{n+1} \leq s_n.$$

Consequently, the nonempty closed intervals $[p_n, s_n]$ are nested.
Step 3: the interval lengths tend to zero. Let $\varepsilon > 0$. By the Cauchy condition, choose N so that $|x_r - x_t| < \varepsilon/3$ whenever $r, t > N$. Fix $n > N$. By the approximation properties of the supremum and infimum, there are indices $r, t \geq n$ such that

$$0 \leq s_n - x_r < \frac{\varepsilon}{3}, \qquad 0 \leq x_t - p_n < \frac{\varepsilon}{3}.$$

Since $r, t > N$, we obtain

$$\begin{aligned} 0 \leq s_n - p_n &\leq |s_n - x_r| + |x_r - x_t| + |x_t - p_n| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

Hence $s_n - p_n \to 0$.
Step 4: convergence to the common point. By lemma 2.57, there is exactly one $x \in \mathbb{R}$ such that

$$\bigcap_{n=1}^{\infty} [p_n, s_n] = \{x\}.$$

Given $\varepsilon > 0$, choose n with $s_n - p_n < \varepsilon$. For every $m \geq n$, both x and x_m belong to $[p_n, s_n]$, so

$$|x_m - x| \leq s_n - p_n < \varepsilon.$$

This proves $x_m \to x$, completing the sufficiency direction. □

2.9. **29.09.2026.**

2.9.1. *Limits of functions.*

Definition 2.59. Let $f : A \to \mathbb{R}$, where $A \subseteq \mathbb{R}$, and let $x_0 \in \mathbb{R}$. Suppose that there is $r > 0$ such that

$$(x_0 - r, x_0 + r) \setminus \{x_0\} \subseteq A.$$

The function f has *limit* $m \in \mathbb{R}$ at x_0 if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|f(x) - m| < \varepsilon \qquad \text{whenever } x \in A \text{ and } 0 < |x - x_0| < \delta.$$

We write $\lim_{x \to x_0} f(x) = m$.

Example 2.60. Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 2x + 1$. Then $\lim_{x \to 1} f(x) = 3$.

Proof. Given $\varepsilon > 0$, choose $\delta = \varepsilon/2$. If $0 < |x - 1| < \delta$, then

$$|f(x) - 3| = |2x - 2| = 2|x - 1| < 2\delta = \varepsilon.$$

□

Example 2.61. Define $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} 2x + 1, & x \neq 1, \\ 42, & x = 1. \end{cases}$$

Again, $\lim_{x\to 1} f(x) = 3$, with exactly the same proof: the condition $0 < |x-1| < \delta$ excludes $x = 1$. Thus the value of a function at the point does not affect its limit.

2.9.2. *Linear estimates and Lipschitz continuity.*

Example 2.62. Suppose that, for some $C > 0$, $r > 0$, and $L \in \mathbb{R}$,

$$|f(x) - L| \leq C|x - x_0| \qquad \text{whenever } 0 < |x - x_0| < r.$$

Then $\lim_{x\to x_0} f(x) = L$. Indeed, given $\varepsilon > 0$, choose $\delta = \min\{r, \varepsilon/C\}$. For $0 < |x-x_0| < \delta$, we have

$$|f(x) - L| \leq C|x - x_0| < C\delta \leq \varepsilon.$$

If the estimate holds throughout the domain, we may simply take $\delta = \varepsilon/C$.

Definition 2.63. A function $f : A \to \mathbb{R}$ is *Lipschitz continuous* if there exists a constant $L \geq 0$ such that

$$|f(x) - f(s)| \leq L|x - s| \qquad \text{for all } x, s \in A.$$

Example 2.64. The function $f : [0, 1] \to \mathbb{R}$, $f(x) = \sqrt{x}$, is not Lipschitz continuous. For every integer $n \geq 1$,

$$\frac{|f(1/n) - f(0)|}{|1/n - 0|} = \frac{\sqrt{1/n}}{1/n} = \sqrt{n} \longrightarrow +\infty.$$

A Lipschitz constant would have to bound all these quotients, which is impossible.

2.9.3. *Choosing a local bound.*

Example 2.65. Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$. Then $\lim_{x\to 2} f(x) = 4$.

Proof. Factor the difference:

$$|f(x) - 4| = |x + 2||x - 2|.$$

If $|x - 2| < 1$, the triangle inequality gives $|x + 2| \leq |x - 2| + 4 < 5$. Given $\varepsilon > 0$, choose $\delta = \min\{1, \varepsilon/5\}$. Whenever $0 < |x - 2| < \delta$, it follows that

$$|f(x) - 4| = |x + 2||x - 2| < 5\delta \leq \varepsilon.$$

□

Example 2.66 (Positivity near a point with positive limit)**.** Let $f : \mathbb{R} \to \mathbb{R}$ and suppose that $\lim_{x\to x_0} f(x) = m > 0$. Then there exists $r > 0$ such that

$$f(x) > 0 \qquad \text{for every } x \in (x_0 - r, x_0 + r) \setminus \{x_0\}.$$

Proof. Since $m > 0$, take $\varepsilon = m$ in the definition of the limit. There exists $\delta > 0$ such that $|f(x) - m| < m$ whenever $0 < |x - x_0| < \delta$. Hence

$$-m < f(x) - m < m, \qquad \text{so} \qquad 0 < f(x) < 2m.$$

Taking $r = \delta$ proves the claim. □

2.9.4. *Uniqueness of the limit.*

Theorem 2.67. *The limit of a function at a point, if it exists, is unique.*

Proof. Let $f : A \to \mathbb{R}$, and suppose that $(x_0 - r, x_0 + r) \setminus \{x_0\} \subseteq A$ for some $r > 0$. Assume, for a contradiction, that

$$\lim_{x \to x_0} f(x) = a, \qquad \lim_{x \to x_0} f(x) = b, \qquad a \neq b.$$

Set $\varepsilon = |a - b|/2 > 0$. There exist $\delta_1, \delta_2 > 0$ such that

$$0 < |x - x_0| < \delta_1 \implies |f(x) - a| < \varepsilon,$$
$$0 < |x - x_0| < \delta_2 \implies |f(x) - b| < \varepsilon.$$

Let $\delta = \min\{r, \delta_1, \delta_2\} > 0$, and choose $x = x_0 + \delta/2$. Then $x \in A$ and $0 < |x - x_0| < \delta$, so both estimates hold. The triangle inequality now gives

$$|a - b| \leq |f(x) - a| + |f(x) - b| < \varepsilon + \varepsilon = |a - b|,$$

a contradiction. Therefore $a = b$. □

2.10. **01.10.2026.**

2.10.1. *Further examples of function limits.*

Example 2.68 (Constant functions)**.** If $f(x) = a$ for every $x \in \mathbb{R}$, then $\lim_{x \to x_0} f(x) = a$ for every $x_0 \in \mathbb{R}$. Indeed, given $\varepsilon > 0$, choose $\delta = 1$. Then $|f(x) - a| = 0 < \varepsilon$ whenever $0 < |x - x_0| < \delta$.

Example 2.69. For $f(x) = x^3$, we have $\lim_{x \to 2} f(x) = 8$.

Proof. If $|x - 2| < 1$, then $1 < x < 3$, so

$$|x^2 + 2x + 4| < 9 + 6 + 4 = 19.$$

Given $\varepsilon > 0$, choose $\delta = \min\{1, \varepsilon/19\}$. For $0 < |x - 2| < \delta$, factor the difference to obtain

$$|x^3 - 8| = |x - 2||x^2 + 2x + 4| < 19|x - 2| < 19\delta \leq \varepsilon.$$

□

2.10.2. *Polynomial division.* Polynomial division lets us factor differences such as $P(x) - P(a)$, which is useful in direct proofs of limits.

Theorem 2.70 (Polynomial division)**.** *Let P, D be polynomials with real coefficients, with $D \neq 0$. There are unique polynomials Q, R such that*

$$P = DQ + R,$$

where either $R = 0$ or $\deg R < \deg D$. We call Q the quotient *and R the* remainder*.*

Proof. Start with $Q = 0$ and $R = P$. If $R = 0$ or $\deg R < \deg D$, we are done. Otherwise, write the leading terms of R and D as ax^m and bx^d, respectively. Since $b \neq 0$ and $m \geq d$, replace

$$Q \quad \text{by} \quad Q + \frac{a}{b}x^{m-d}, \qquad R \quad \text{by} \quad R - \frac{a}{b}x^{m-d}D.$$

The identity $P = DQ + R$ is preserved, and the leading term of R cancels. Thus the remainder becomes zero or its degree strictly decreases. After finitely many steps the required degree condition holds.

For uniqueness, suppose $DQ + R = D\widetilde{Q} + \widetilde{R}$, with both remainders satisfying the degree condition. Then

$$D(Q - \widetilde{Q}) = \widetilde{R} - R.$$

If $Q - \widetilde{Q} \neq 0$, the left side has degree at least $\deg D$, while the right side is zero or has smaller degree. This is impossible. Hence $Q = \widetilde{Q}$ and $R = \widetilde{R}$. □

Corollary 2.71 (Remainder and factor theorems)**.** *The remainder when dividing $P(x)$ by $x-a$ is $P(a)$. In particular, $x-a$ is a factor of $P(x)$ if and only if $P(a)=0$.*

Proof. Polynomial division gives $P(x) = (x-a)Q(x) + r$ with constant remainder r. Substituting $x = a$ gives $r = P(a)$. Thus the remainder vanishes exactly when $P(a) = 0$. □

To divide by hand, arrange terms in descending powers, insert zero coefficients for missing powers, and repeatedly divide the leading terms, multiply the divisor, and subtract.

Example 2.72 (Cancelling a factor in a limit)**.** Consider the limit

$$\lim_{x\to 2} \frac{x^3+5x^2-2x-24}{x-2}.$$

Both numerator and denominator tend to zero, so direct use of the quotient law is not possible. Divide $P(x) = x^3+5x^2-2x-24$ by $x-2$. The successive quotient terms are x^2, $7x$, and 12:

$$\begin{aligned} P(x) - x^2(x-2) &= 7x^2 - 2x - 24, \\ (7x^2-2x-24) - 7x(x-2) &= 12x - 24, \\ (12x-24) - 12(x-2) &= 0. \end{aligned}$$

Therefore

$$x^3+5x^2-2x-24 = (x-2)(x^2+7x+12).$$

For $x \neq 2$, we can cancel $x-2$, giving

$$\frac{x^3+5x^2-2x-24}{x-2} = x^2+7x+12.$$

Thus the desired limit is $4+14+12 = 30$. A direct verification uses

$$|x^2+7x+12-30| = |x-2||x+9|.$$

If $|x-2|<1$, then $|x+9|<12$. Given $\varepsilon>0$, take $\delta = \min\{1, \varepsilon/12\}$. For $0<|x-2|<\delta$, the difference from 30 is less than $12\delta \leq \varepsilon$. The quotient need not be defined at $x=2$, because the limit uses only nearby points.

Example 2.73 (Using a remainder to simplify a limit)**.** Consider

$$\lim_{x\to 1} \frac{x^3-1}{x^2-1}.$$

Again, both numerator and denominator tend to zero. Dividing x^3-1 by x^2-1 gives quotient x and remainder $x-1$, since

$$(x^3-1) - x(x^2-1) = x-1.$$

The remainder has degree $1<2$, so division is complete. For $x \neq \pm 1$, factor the denominator and cancel:

$$\frac{x^3-1}{x^2-1} = x + \frac{x-1}{x^2-1} = x + \frac{x-1}{(x-1)(x+1)} = x + \frac{1}{x+1}.$$

This suggests the limit $1+1/2 = 3/2$. To verify it directly, suppose $0<|x-1|<1/2$. Then $x+1>3/2$, and

$$\begin{aligned} \left|\frac{x^3-1}{x^2-1} - \frac{3}{2}\right| &= \left|(x-1) + \frac{1}{x+1} - \frac{1}{2}\right| \\ &\leq |x-1| + \frac{|x-1|}{2|x+1|} \\ &< 2|x-1|. \end{aligned}$$

Given $\varepsilon>0$, take $\delta = \min\{1/2, \varepsilon/2\}$. The displayed estimate is then less than $2\delta \leq \varepsilon$, proving the limit.

Remark 2.74. In both examples, division exposes a common factor that can be cancelled for $x \neq x_0$. Functions that agree throughout a punctured neighbourhood of x_0 have the same limit there: their defining ε–δ estimates are identical after shrinking δ to stay in that neighbourhood.

2.10.3. *Limits using factorisation and local estimates.*

Example 2.75. For $f(x) = x^3 + 5x^2 - 2x - 24$, we have $\lim_{x\to 2} f(x) = 0$.

Proof. By example 2.72, $f(x) = (x-2)(x^2+7x+12)$. If $|x-2| < 1$, then $1 < x < 3$, and hence

$$|x^2 + 7x + 12| < 9 + 21 + 12 = 42.$$

Given $\varepsilon > 0$, choose $\delta = \min\{1, \varepsilon/42\}$. If $0 < |x-2| < \delta$, then

$$|f(x) - 0| < 42|x-2| < 42\delta \leq \varepsilon.$$

□

Example 2.76 (A piecewise function)**.** Define $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} 2|x|, & x < 0, \\ 7, & x = 0, \\ x^2, & x > 0. \end{cases}$$

Then $\lim_{x\to 0} f(x) = 0$, even though $f(0) = 7$.

Proof. Given $\varepsilon > 0$, take $\delta = \min\{1, \varepsilon/2\}$. If $-\delta < x < 0$, then $|f(x)| = 2|x| < 2\delta \leq \varepsilon$. If $0 < x < \delta$, then $x < 1$, so $|f(x)| = x^2 < x < \delta \leq \varepsilon/2 < \varepsilon$. These cases cover $0 < |x| < \delta$. □

Example 2.77 (Keeping a denominator away from zero)**.** Let $f : (0,1) \to \mathbb{R}$, $f(x) = 7/x$. Then $\lim_{x\to 1/2} f(x) = 14$.

Proof. We have

$$\left|\frac{7}{x} - 14\right| = \frac{14}{|x|}\left|x - \frac{1}{2}\right|.$$

If $|x - 1/2| < 1/4$, then $x > 1/4$, so $14/|x| < 56$. Given $\varepsilon > 0$, choose $\delta = \min\{1/4, \varepsilon/56\}$. Then

$$0 < \left|x - \frac{1}{2}\right| < \delta \quad \Longrightarrow \quad |f(x) - 14| < 56\left|x - \frac{1}{2}\right| < 56\delta \leq \varepsilon.$$

□

2.10.4. *A necessary Cauchy condition for function limits.* In the following results, domains contain a punctured neighbourhood of x_0: for some $r > 0$, $(x_0 - r, x_0 + r) \setminus \{x_0\} \subseteq A$, as in the previous lecture's definition. All points at which a function is evaluated are understood to belong to its domain.

Theorem 2.78. *If $\lim_{x\to x_0} f(x) = L \in \mathbb{R}$, then for every $\varepsilon > 0$ there exists $\delta > 0$ such that*

$$|f(x) - f(y)| < \varepsilon$$

whenever $0 < |x - x_0| < \delta$ and $0 < |y - x_0| < \delta$.

Proof. Choose $\delta > 0$ so that $|f(t) - L| < \varepsilon/2$ whenever $0 < |t - x_0| < \delta$. For the stated x, y,

$$|f(x) - f(y)| \leq |f(x) - L| + |f(y) - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

□

Example 2.79 (A jump prevents a limit)**.** Let $f(x) = 1$ for $x \leq 0$, and $f(x) = 2$ for $x > 0$. The limit at 0 does not exist. Indeed, fix $\varepsilon = 1/2$. For every $\delta > 0$, the points $x = \delta/2$ and $y = -\delta/2$ satisfy $0 < |x|, |y| < \delta$, but

$$|f(x) - f(y)| = 1 > \varepsilon.$$

This contradicts the necessary condition in theorem 2.78.

2.10.5. *The sequential criterion for function limits.*

Theorem 2.80 (Sequential criterion)**.** *Let $f : A \to \mathbb{R}$ and $m \in \mathbb{R}$. The following are equivalent:*

(a) $\lim_{x \to x_0} f(x) = m$.

(b) For every sequence (y_n) in $A \setminus \{x_0\}$ with $y_n \to x_0$, we have $f(y_n) \to m$.

Proof. Suppose (a) holds, and let $y_n \to x_0$ with $y_n \in A \setminus \{x_0\}$. Given $\varepsilon > 0$, choose $\delta > 0$ such that

$$0 < |x - x_0| < \delta \implies |f(x) - m| < \varepsilon.$$

There is $N \in \mathbb{N}$ such that $|y_n - x_0| < \delta$ for $n > N$. Since $y_n \neq x_0$, it follows that $|f(y_n) - m| < \varepsilon$ for $n > N$. Thus (b) holds.

Conversely, suppose (b) holds but (a) fails. The negation of (a) says that there exists $\varepsilon_0 > 0$ such that for every $\delta > 0$ we can find $y \in A$ satisfying

$$0 < |y - x_0| < \delta, \qquad |f(y) - m| \geq \varepsilon_0.$$

For each $n \in \mathbb{N}$, choose such a point y_n with $\delta = \min\{r, 1/(n+1)\}$. Then $y_n \neq x_0$ and $|y_n - x_0| < 1/(n+1)$, so $y_n \to x_0$. But $|f(y_n) - m| \geq \varepsilon_0$ for every n, contradicting (b). Therefore (a) holds. □

Remark 2.81. In the converse proof, we negate the statement that the limit equals the specified number m. This includes both a nonexistent limit and a limit different from m. The restriction $y_n \neq x_0$ is essential because $f(x_0)$ need not equal the limit.

2.10.6. *Local boundedness and preservation of bounds.*

Theorem 2.82 (Local boundedness)**.** *If $\lim_{x \to x_0} f(x) = L \in \mathbb{R}$, there are $M, \delta > 0$ such that $|f(x)| < M$ whenever $0 < |x - x_0| < \delta$.*

Proof. Choose $\delta > 0$ using $\varepsilon = 1$ in the limit definition. Then

$$|f(x)| \leq |f(x) - L| + |L| < 1 + |L|$$

whenever $0 < |x - x_0| < \delta$. Take $M = 1 + |L| > 0$. □

Theorem 2.83 (Preservation of bounds)**.** *Suppose $\lim_{x \to x_0} f(x) = L \in \mathbb{R}$. If $f(x) \leq M$ throughout some punctured neighbourhood of x_0, then $L \leq M$. If $f(x) \geq m$ throughout such a neighbourhood, then $L \geq m$. In particular, these conclusions hold if the bounds hold on all of A.*

Proof. For the upper bound, suppose $L > M$ and choose $\varepsilon = L - M > 0$. The limit definition gives $\delta > 0$ such that

$$f(x) > L - \varepsilon = M \qquad (0 < |x - x_0| < \delta).$$

Shrink δ if necessary so that this neighbourhood lies in A and the assumed bound holds there. Taking $x = x_0 + \delta/2$ contradicts $f(x) \leq M$. Hence $L \leq M$.

For the lower bound, suppose $L < m$ and take $\varepsilon = m - L > 0$. For all sufficiently close $x \neq x_0$,

$$f(x) < L + \varepsilon = m.$$

Choosing such a point in the neighbourhood where $f(x) \geq m$ gives a contradiction. Thus $L \geq m$. □

2.10.7. *Arithmetic laws for function limits.*

Theorem 2.84. *Let $f, g : A \to \mathbb{R}$, with $\lim_{x\to x_0} f(x) = a$ and $\lim_{x\to x_0} g(x) = b$. Then, for every $c \in \mathbb{R}$,*

$$\lim_{x\to x_0} (f(x) + g(x)) = a + b,$$
$$\lim_{x\to x_0} cf(x) = ca,$$
$$\lim_{x\to x_0} f(x)g(x) = ab.$$

If $b \neq 0$, then $g(x) \neq 0$ in some punctured neighbourhood of x_0, and

$$\lim_{x\to x_0} \frac{f(x)}{g(x)} = \frac{a}{b}.$$

The quotient is defined on $\{x \in A : g(x) \neq 0\}$.

Proof. Let $\varepsilon > 0$. All estimates below apply to points in a sufficiently small punctured neighbourhood of x_0.
Sum. Choose $\delta_f, \delta_g > 0$ so that $|f(x) - a| < \varepsilon/2$ for $0 < |x - x_0| < \delta_f$, and $|g(x) - b| < \varepsilon/2$ for $0 < |x - x_0| < \delta_g$. With $\delta = \min\{\delta_f, \delta_g\}$,

$$|f(x) + g(x) - (a + b)| \leq |f(x) - a| + |g(x) - b| < \varepsilon.$$

Scalar multiple. If $c = 0$, the result follows from the constant-function example. If $c \neq 0$, choose $\delta > 0$ so that $|f(x) - a| < \varepsilon/|c|$. Then

$$|cf(x) - ca| = |c||f(x) - a| < \varepsilon.$$

Product. By theorem 2.82, there are $M > 0$ and $\delta_0 > 0$ such that $|f(x)| < M$ when $0 < |x - x_0| < \delta_0$. Choose $\delta_1, \delta_2 > 0$ to ensure respectively that

$$|g(x) - b| < \frac{\varepsilon}{2M}, \qquad |f(x) - a| < \frac{\varepsilon}{2(|b| + 1)}.$$

For $0 < |x - x_0| < \min\{\delta_0, \delta_1, \delta_2\}$,

$$\begin{aligned} |f(x)g(x) - ab| &= |f(x)(g(x) - b) + b(f(x) - a)| \\ &\leq |f(x)||g(x) - b| + |b||f(x) - a| \\ &< \frac{\varepsilon}{2} + \frac{|b|\varepsilon}{2(|b| + 1)} < \varepsilon. \end{aligned}$$

Quotient. Assume $b \neq 0$. First choose $\delta_0 > 0$ such that $|g(x) - b| < |b|/2$. The reverse triangle inequality gives

$$|g(x)| \geq |b| - |g(x) - b| > \frac{|b|}{2} > 0.$$

Thus the quotient is defined near x_0. To prove $1/g(x) \to 1/b$, choose $\delta_1 > 0$ so that $|g(x) - b| < \varepsilon|b|^2/2$. For $0 < |x - x_0| < \min\{\delta_0, \delta_1\}$,

$$\left|\frac{1}{g(x)} - \frac{1}{b}\right| = \frac{|g(x) - b|}{|g(x)||b|} \leq \frac{2|g(x) - b|}{|b|^2} < \varepsilon.$$

The product law applied to f and $1/g$ now yields $f(x)/g(x) \to a/b$. □

Corollary 2.85 (Polynomial and rational limits)**.** *For every real polynomial P and every $x_0 \in \mathbb{R}$, $\lim_{x\to x_0} P(x) = P(x_0)$. If Q is another polynomial with $Q(x_0) \neq 0$, then*

$$\lim_{x\to x_0} \frac{P(x)}{Q(x)} = \frac{P(x_0)}{Q(x_0)}.$$

Proof. The identity function has limit x_0 at x_0: choose $\delta = \varepsilon$. Repeated use of the product law gives $x^k \to x_0^k$ for each positive integer k. Constant functions, scalar multiples, and finite sums then give the polynomial statement. The rational statement follows from the quotient law. □

2.11. **06.10.2026.**

2.11.1. *Infinite limits at a finite point.* As before, assume that the domain of $f : A \to \mathbb{R}$ contains a punctured neighbourhood of $x_0 \in \mathbb{R}$.

Definition 2.86. We write $\lim_{x\to x_0} f(x) = +\infty$ if for every $M \in \mathbb{R}$ there is $\delta > 0$ such that

$$0 < |x - x_0| < \delta \quad \Longrightarrow \quad f(x) > M.$$

Similarly, $\lim_{x\to x_0} f(x) = -\infty$ if for every $M \in \mathbb{R}$ there is $\delta > 0$ such that

$$0 < |x - x_0| < \delta \quad \Longrightarrow \quad f(x) < M.$$

Example 2.87. For $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$, $f(x) = 1/|x|$, we have $\lim_{x\to 0} f(x) = +\infty$. If $M \le 0$, every $x \neq 0$ satisfies $f(x) > M$. If $M > 0$, choose $\delta = 1/M$. Then $0 < |x| < \delta$ implies $f(x) = 1/|x| > 1/\delta = M$.

2.11.2. *One-sided limits.* For the following definitions, the domain A need only contain the corresponding one-sided interval near x_0.

Definition 2.88. The function $f : A \to \mathbb{R}$ has *left-hand limit* $L \in \mathbb{R}$ at x_0, written $\lim_{x\to x_0^-} f(x) = L$, if for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$x_0 - \delta < x < x_0 \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

The *right-hand limit* $\lim_{x\to x_0^+} f(x) = L$ is defined by requiring

$$x_0 < x < x_0 + \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

The left-hand limit is also denoted by $\lim_{x\to x_0-} f(x)$, and similarly on the right.

Example 2.89. Let

$$f(x) = \begin{cases} 1 - x, & x \le 0, \\ 2x - 1, & x > 0. \end{cases}$$

For $x < 0$, $|f(x) - 1| = |x|$, so choosing $\delta = \varepsilon$ proves $\lim_{x\to 0^-} f(x) = 1$. For $x > 0$, $|f(x) - (-1)| = 2|x|$, so choosing $\delta = \varepsilon/2$ proves $\lim_{x\to 0^+} f(x) = -1$. The two-sided limit at 0 therefore does not exist.

Theorem 2.90. *Suppose A contains a punctured neighbourhood of x_0. Then $\lim_{x\to x_0} f(x) = L \in \mathbb{R}$ if and only if*

$$\lim_{x\to x_0^-} f(x) = \lim_{x\to x_0^+} f(x) = L.$$

Proof. If the two-sided limit is L, its δ for a given $\varepsilon > 0$ works on each side of x_0.

Conversely, suppose both one-sided limits equal L. Given $\varepsilon > 0$, choose δ_- and δ_+ from the respective definitions, and put $\delta = \min\{\delta_-, \delta_+\}$. If $0 < |x - x_0| < \delta$, then x is either in $(x_0 - \delta_-, x_0)$ or in $(x_0, x_0 + \delta_+)$. In either case $|f(x) - L| < \varepsilon$. □

Example 2.91. For $f : [0, \infty) \to \mathbb{R}$, $f(x) = \sqrt{x}$, we have $\lim_{x\to 0^+} f(x) = 0$. Given $\varepsilon > 0$, take $\delta = \varepsilon^2$. Then $0 < x < \delta$ implies $|f(x) - 0| = \sqrt{x} < \sqrt{\delta} = \varepsilon$. Here only the right-hand limit is defined, since the domain has no points immediately to the left of 0.

2.11.3. *A necessary Cauchy condition for one-sided limits.*

Theorem 2.92. *Let $(x_0, x_0 + r) \subseteq A$ for some $r > 0$, and suppose $\lim_{x \to x_0^+} f(x) = L \in \mathbb{R}$. Then for every $\varepsilon > 0$ there is $\delta > 0$ such that*

$$|f(x) - f(y)| < \varepsilon \qquad \text{whenever } x, y \in (x_0, x_0 + \delta).$$

The analogous statement holds for a left-hand limit.

Proof. Choose $0 < \delta \leq r$ so that $|f(t) - L| < \varepsilon/2$ for all $t \in (x_0, x_0 + \delta)$. Then, for any x, y in this interval,

$$|f(x) - f(y)| \leq |f(x) - L| + |f(y) - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

The left-hand case is identical. □

2.12. **09.10.2026.**

2.12.1. *Oscillation and one-sided limits.*

Example 2.93 (An oscillating function without a right-hand limit)**.** Let $f : (0, \infty) \to \mathbb{R}$, $f(x) = \sin(1/x)$. Then $\lim_{x \to 0^+} f(x)$ does not exist.

Proof. For $n \in \mathbb{N}$, put

$$x_n = \frac{1}{\pi/2 + 2\pi n}, \qquad y_n = \frac{1}{3\pi/2 + 2\pi n}.$$

Both sequences are positive and tend to zero. Indeed, for $n \geq 1$, $0 < x_n < 1/n$ and $0 < y_n < 1/n$. However,

$$f(x_n) = 1, \qquad f(y_n) = -1, \qquad |f(x_n) - f(y_n)| = 2.$$

For every $\delta > 0$, sufficiently large n therefore gives $x_n, y_n \in (0, \delta)$ while their function values remain distance 2 apart. Taking $\varepsilon = 1$ contradicts the necessary Cauchy condition for a finite right-hand limit. The function cannot tend to either infinity either, since $-1 \leq f(x) \leq 1$. □

Example 2.94 (Damping the oscillation)**.** If $g : (0, \infty) \to \mathbb{R}$ satisfies $\lim_{x \to 0^+} g(x) = 0$, then

$$\lim_{x \to 0^+} g(x) \sin(1/x) = 0.$$

Proof. Since $|\sin(1/x)| \leq 1$, we have

$$|g(x) \sin(1/x) - 0| \leq |g(x)|.$$

Given $\varepsilon > 0$, choose $\delta > 0$ such that $0 < x < \delta$ implies $|g(x)| < \varepsilon$. The displayed inequality proves the claimed limit. □

Remark 2.95. The usual product law cannot be applied directly here, because $\sin(1/x)$ has no right-hand limit at zero. The estimate works because one factor tends to zero and the other is bounded.

2.12.2. *The indicator function of the rational numbers.*

Example 2.96. Define $\chi_{\mathbb{Q}} : \mathbb{R} \to \mathbb{R}$ by

$$\chi_{\mathbb{Q}}(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Then $\lim_{x \to 0} \chi_{\mathbb{Q}}(x)$ does not exist.

Proof directly from the definition. Suppose that the limit is $L \in \mathbb{R}$. With $\varepsilon = 1/2$, there is $\delta > 0$ such that

$$0 < |x| < \delta \implies |\chi_{\mathbb{Q}}(x) - L| < \frac{1}{2}.$$

Choose a positive integer $n > \sqrt{2}/\delta$. Then $u = 1/n$ and $v = \sqrt{2}/n$ both lie in $(0, \delta)$. The number u is rational and v is irrational: if v were rational, then $nv = \sqrt{2}$ would be rational as well. Thus $|1 - L| < 1/2$ and $|L| < 1/2$, giving

$$1 \leq |1 - L| + |L| < 1,$$

a contradiction. □

Proof using sequences. For positive integers n, both $1/n$ and $\sqrt{2}/n$ tend to zero, but

$$\chi_{\mathbb{Q}}(1/n) = 1, \qquad \chi_{\mathbb{Q}}(\sqrt{2}/n) = 0.$$

By theorem 2.80, a limit at zero would have to equal both 1 and 0, which is impossible. □

2.12.3. *Arithmetic laws for one-sided limits.*

Theorem 2.97. *Let $f, g : A \to \mathbb{R}$, and suppose $(x_0 - r, x_0) \subseteq A$ for some $r > 0$. If*

$$\lim_{x \to x_0^-} f(x) = a, \qquad \lim_{x \to x_0^-} g(x) = b,$$

then, for every $c \in \mathbb{R}$,

$$\lim_{x \to x_0^-} (f(x) + g(x)) = a + b,$$
$$\lim_{x \to x_0^-} cf(x) = ca,$$
$$\lim_{x \to x_0^-} f(x)g(x) = ab.$$

If $b \neq 0$, then $g(x) \neq 0$ sufficiently close to x_0 on the left, and

$$\lim_{x \to x_0^-} \frac{f(x)}{g(x)} = \frac{a}{b}.$$

The quotient is defined where $g(x) \neq 0$. The same laws hold for right-hand limits when $(x_0, x_0 + r) \subseteq A$.

Proof. Let $\varepsilon > 0$. All radii below are chosen at most r, so the relevant intervals lie in A. Sum. Choose $\delta_f, \delta_g > 0$ so that $|f(x) - a| < \varepsilon/2$ for $x_0 - \delta_f < x < x_0$, and $|g(x) - b| < \varepsilon/2$ for $x_0 - \delta_g < x < x_0$. With $\delta = \min\{\delta_f, \delta_g\}$, for $x_0 - \delta < x < x_0$ we obtain

$$|f(x) + g(x) - (a + b)| \leq |f(x) - a| + |g(x) - b| < \varepsilon.$$

Scalar multiple. If $c = 0$, the difference $cf(x) - ca$ is identically zero. Otherwise choose $\delta > 0$ so that $|f(x) - a| < \varepsilon/|c|$ for $x_0 - \delta < x < x_0$. Then $|cf(x) - ca| = |c||f(x) - a| < \varepsilon$. Product. First choose $\delta_0 > 0$ such that $|g(x) - b| < 1$ on $(x_0 - \delta_0, x_0)$. There, $|g(x)| < |b| + 1$. Choose $\delta_a, \delta_b > 0$ so that, on the respective left-hand intervals,

$$|f(x) - a| < \frac{\varepsilon}{2(|b| + 1)}, \qquad |g(x) - b| < \frac{\varepsilon}{2(|a| + 1)}.$$

Set $\delta = \min\{\delta_0, \delta_a, \delta_b\}$. For $x_0 - \delta < x < x_0$,

$$\begin{aligned} |f(x)g(x) - ab| &= |g(x)(f(x) - a) + a(g(x) - b)| \\ &\leq |g(x)||f(x) - a| + |a||g(x) - b| \\ &\leq (|b| + 1)|f(x) - a| + |a||g(x) - b| \\ &< \frac{\varepsilon}{2} + \frac{|a|\varepsilon}{2(|a| + 1)} < \varepsilon. \end{aligned}$$

Quotient. Suppose $b \neq 0$. Choose $\delta_0 > 0$ so that $|g(x) - b| < |b|/2$ for $x_0 - \delta_0 < x < x_0$. Then

$$|g(x)| \geq |b| - |g(x) - b| > \frac{|b|}{2} > 0.$$

Choose $\delta_1 > 0$ so that $|g(x) - b| < \varepsilon |b|^2/2$ for $x_0 - \delta_1 < x < x_0$. On the interval with radius $\min\{\delta_0, \delta_1\}$,

$$\left| \frac{1}{g(x)} - \frac{1}{b} \right| = \frac{|g(x) - b|}{|g(x)||b|} \leq \frac{2|g(x) - b|}{|b|^2} < \varepsilon.$$

Thus $1/g(x) \to 1/b$ from the left. Applying the product law just proved to f and $1/g$ proves the quotient law.

For right-hand limits, replace each interval $(x_0 - \delta, x_0)$ by $(x_0, x_0 + \delta)$. The same choices and estimates prove all four statements. □

2.12.4. *Finite limits at infinity.*

Definition 2.98. Let $f : A \to \mathbb{R}$, and suppose that $(r, \infty) \subseteq A$ for some $r \in \mathbb{R}$. We write $\lim_{x \to +\infty} f(x) = L \in \mathbb{R}$ if for every $\varepsilon > 0$ there exists $R \geq r$ such that

$$x > R \implies |f(x) - L| < \varepsilon.$$

Similarly, if $(-\infty, r) \subseteq A$, we write $\lim_{x \to -\infty} f(x) = L$ if for every $\varepsilon > 0$ there exists $R \leq r$ such that

$$x < R \implies |f(x) - L| < \varepsilon.$$

Example 2.99. For $f : (1, \infty) \to \mathbb{R}$, $f(x) = 1/x$, we have $\lim_{x \to +\infty} f(x) = 0$.

Proof. Given $\varepsilon > 0$, take $R = \max\{1, 1/\varepsilon\}$. For $x > R$,

$$|f(x) - 0| = \frac{1}{x} < \varepsilon.$$

□

Example 2.100. For $f : (3, \infty) \to \mathbb{R}$, $f(x) = 2x/(x - 3)$, we have $\lim_{x \to +\infty} f(x) = 2$.

Proof. Given $\varepsilon > 0$, take $R = \max\{6, 12/\varepsilon\}$. If $x > R$, then $x/2 - 3 > 0$, and hence

$$\left| \frac{2x}{x - 3} - 2 \right| = \frac{6}{x - 3} = \frac{6}{x/2 + (x/2 - 3)} < \frac{12}{x} < \varepsilon.$$

□

Example 2.101. For $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x/(|x| + 1)$, we have $\lim_{x \to -\infty} f(x) = -1$.

Proof. Given $\varepsilon > 0$, take $R = \min\{0, -1/\varepsilon\}$. If $x < R$, then $x < 0$, so $|x| = -x$. Therefore

$$\left| \frac{x}{|x| + 1} - (-1) \right| = \left| \frac{x + |x| + 1}{|x| + 1} \right| = \frac{1}{1 - x} < \frac{1}{-x} < \varepsilon.$$

□

2.12.5. *Infinite limits at infinity.*

Definition 2.102. Suppose $(r, \infty) \subseteq A$. We write $\lim_{x \to +\infty} f(x) = +\infty$ if for every $M \in \mathbb{R}$ there exists $R \geq r$ such that $x > R$ implies $f(x) > M$. We write $\lim_{x \to +\infty} f(x) = -\infty$ if instead $x > R$ implies $f(x) < M$.

If $(-\infty, r) \subseteq A$, the definitions at $-\infty$ use $R \leq r$ and $x < R$: for every $M \in \mathbb{R}$, eventually $f(x) > M$ for a limit of $+\infty$, or eventually $f(x) < M$ for a limit of $-\infty$.

Example 2.103. For $f : (-\infty, -3) \to \mathbb{R}$, $f(x) = x^2/(x + 3)$, we have

$$\lim_{x \to -\infty} f(x) = -\infty.$$

Proof. Let $M \in \mathbb{R}$, and choose $R = \min\{-3, M\}$. For $x < R$, we have $x < x + 3 < 0$, and consequently $1/(x+3) < 1/x$. Multiplication by $x^2 > 0$ gives

$$f(x) = \frac{x^2}{x+3} < \frac{x^2}{x} = x < M.$$

This is precisely the definition of the asserted infinite limit. □

2.13. **13.10.2026.**

2.13.1. *Monotone functions.*

Definition 2.104. Let $A \subseteq \mathbb{R}$ and $f : A \to \mathbb{R}$. The function f is

(a) *increasing* on A if $x_1 \le x_2$ implies $f(x_1) \le f(x_2)$;
(b) *strictly increasing* on A if $x_1 < x_2$ implies $f(x_1) < f(x_2)$;
(c) *decreasing* on A if $x_1 \le x_2$ implies $f(x_1) \ge f(x_2)$;
(d) *strictly decreasing* on A if $x_1 < x_2$ implies $f(x_1) > f(x_2)$;
(e) *monotone* on A if it is increasing or decreasing;
(f) *strictly monotone* on A if it is strictly increasing or strictly decreasing.

In each implication, $x_1, x_2 \in A$. Increasing and decreasing in the non-strict sense are also called *nondecreasing* and *nonincreasing*, respectively.

Theorem 2.105. *Every strictly monotone function is injective.*

Proof. Let $f : A \to \mathbb{R}$ be strictly monotone, and let $x_1, x_2 \in A$ be distinct. Interchanging the points if necessary, assume $x_1 < x_2$. If f is strictly increasing, then $f(x_1) < f(x_2)$; if it is strictly decreasing, then $f(x_1) > f(x_2)$. In either case $f(x_1) \neq f(x_2)$, which proves injectivity. □

Example 2.106. For every positive integer n, the function

$$f_n : [0, \infty) \to \mathbb{R}, \qquad f_n(x) = x^n,$$

is strictly increasing, and hence injective.

Proof. We use induction on n. For $n = 1$, the assertion follows from $f_1(x) = x$. Suppose that f_n is strictly increasing for some $n \ge 1$, and let $0 \le x_1 < x_2$. Since $x_1 \ge 0$ and $x_2^n > 0$,

$$x_1^{n+1} = x_1 x_1^n \le x_1 x_2^n < x_2 x_2^n = x_2^{n+1}.$$

Thus f_{n+1} is strictly increasing, completing the induction. □

2.13.2. *Endpoint limits of monotone functions.*

Theorem 2.107. *Let $a, b \in \mathbb{R}$, $a < b$, and let $f : (a, b) \to \mathbb{R}$ be increasing.*

(a) If $f((a,b))$ is bounded below, then

$$\lim_{x \to a^+} f(x) = \inf\{f(x) : x \in (a,b)\} \in \mathbb{R}.$$

(b) If $f((a,b))$ is bounded above, then

$$\lim_{x \to b^-} f(x) = \sup\{f(x) : x \in (a,b)\} \in \mathbb{R}.$$

Proof. For (a), let $p = \inf f((a,b))$. The image is nonempty and bounded below, so completeness gives $p \in \mathbb{R}$. Given $\varepsilon > 0$, the definition of the infimum gives a point $x_p \in (a,b)$ such that

$$p \le f(x_p) < p + \varepsilon.$$

Indeed, otherwise $p + \varepsilon$ would be a lower bound, contradicting the choice of p. By monotonicity,

$$p \le f(x) \le f(x_p) < p + \varepsilon \qquad (a < x < x_p).$$

Choose $\delta = x_p - a > 0$. For $a < x < a + \delta$, we have $|f(x) - p| < \varepsilon$, proving the right-hand limit.

For (b), let $s = \sup f((a,b)) \in \mathbb{R}$. Given $\varepsilon > 0$, choose $x_s \in (a,b)$ such that

$$s - \varepsilon < f(x_s) \leq s.$$

Such a point exists because $s - \varepsilon$ is not an upper bound. Monotonicity now gives

$$s - \varepsilon < f(x_s) \leq f(x) \leq s \qquad (x_s < x < b).$$

With $\delta = b - x_s > 0$, it follows that $|f(x) - s| < \varepsilon$ for $b - \delta < x < b$. This proves the left-hand limit. □

Corollary 2.108. *Let $f : (a,b) \to \mathbb{R}$ be decreasing, where $a < b$ are real. If $f((a,b))$ is bounded above, then*

$$\lim_{x \to a^+} f(x) = \sup f((a,b)) \in \mathbb{R}.$$

If $f((a,b))$ is bounded below, then

$$\lim_{x \to b^-} f(x) = \inf f((a,b)) \in \mathbb{R}.$$

Proof. The function $g = -f$ is increasing. If $f((a,b))$ is bounded above, then $g((a,b))$ is bounded below, and theorem 2.107 gives

$$\lim_{x \to a^+} g(x) = \inf g((a,b)) = -\sup f((a,b)).$$

Multiplying by -1 proves the first assertion. If $f((a,b))$ is bounded below, then $g((a,b))$ is bounded above, so the same theorem gives

$$\lim_{x \to b^-} g(x) = \sup g((a,b)) = -\inf f((a,b)).$$

Multiplying by -1 proves the second assertion. □

2.13.3. *One-sided limits on a closed interval.*

Theorem 2.109. *Let $f : [a,b] \to \mathbb{R}$ be monotone, where $a < b$. Then f has finite left-hand and right-hand limits at every $x_0 \in (a,b)$, as well as finite limits from within the interval at a and b.*

If f is increasing, then

$$\lim_{x \to x_0^-} f(x) \leq f(x_0) \leq \lim_{x \to x_0^+} f(x),$$

and

$$f(a) \leq \lim_{x \to a^+} f(x), \qquad \lim_{x \to b^-} f(x) \leq f(b).$$

If f is decreasing, all these inequalities are reversed.

Proof. First suppose that f is increasing and fix $x_0 \in (a,b)$. The sets

$$A = \{f(x) : a < x < x_0\}, \qquad B = \{f(x) : x_0 < x < b\}$$

are nonempty. For $a < x < x_0$, monotonicity gives $f(x) \leq f(x_0)$, so A is bounded above by $f(x_0)$. Applying theorem 2.107 to the restriction of f to (a, x_0), we obtain

$$\lim_{x \to x_0^-} f(x) = \sup A \leq f(x_0).$$

For $x_0 < x < b$, we have $f(x_0) \leq f(x)$, so B is bounded below by $f(x_0)$. Applying the same theorem on (x_0, b) gives

$$\lim_{x \to x_0^+} f(x) = \inf B \geq f(x_0).$$

Both limits are finite by completeness.

For the endpoint limits, observe that

$$f(a) \le f(x) \le f(b) \qquad (a < x < b).$$

Thus $f((a,b))$ is bounded both below and above. The endpoint theorem yields

$$\lim_{x\to a^+} f(x) = \inf f((a,b)) \ge f(a), \qquad \lim_{x\to b^-} f(x) = \sup f((a,b)) \le f(b).$$

If f is decreasing, then $g = -f$ is increasing. Applying the result just proved to g establishes all the required finite limits. Its inequalities at an interior point become

$$-\lim_{x\to x_0^-} f(x) \le -f(x_0) \le -\lim_{x\to x_0^+} f(x).$$

Multiplying by -1 reverses the inequalities and gives

$$\lim_{x\to x_0^-} f(x) \ge f(x_0) \ge \lim_{x\to x_0^+} f(x).$$

The endpoint inequalities for g likewise give $f(a) \ge \lim_{x\to a^+} f(x)$ and $\lim_{x\to b^-} f(x) \ge f(b)$. □